

Fig. 3: Actual-size PCB layout of GPS- and GSM-based vehicle tracking system

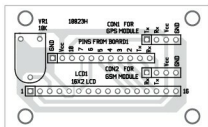


Fig. 4: Components layout for the PCB

Once you are done with connections and uploading of the code, place the system (Arduino, LCD, GPS and GSM with SIM card) in the vehicle. Now it's time to track the SIM card inserted in the GSM module. To track the SIM card,

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno
- Resistors (all 1/4-watt, $\pm 5\%$ carbon):
- VR1 - 10-kilo-ohm preset

Miscellaneous:

- LCD1 - 16 $\times$ 2 LCD
- CON1 - 4-pin connector for GPS module
- CON2 - 4-pin connector for GSM module
- 9V battery for Board1
- Jumper wires
- GSM SIM card
- Jack1 - DC jack
- NEO-6M GPS module with antenna
- GSM module SIM900

send an SMS to the mobile number of the SIM card by typing "Track Vehicle" from your (user) mobile phone.

After sending the message, wait for a few seconds until you receive location detail of the SIM card in the form of latitude and longitude values on LCD1 as well as in your mobile phone. Vary preset VR1 till you get clear visible text on the LCD1. After getting the latitude and longitude position (in decimal format), convert the first four digits of these values into degrees and minutes

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

like 28°38' and 77°13'. Enter these values in the search box of Google Maps and then click Search button to find the exact location.

Applications

The system can be used by individual vehicle owners as well as organisations to locate and track their vehicles. Organisations can use it to track the status of their product delivery by trucks at all times. It also helps owners to keep tab on their driver, preventing vehicle abuse and thus resulting in significant cost-savings. The system can be easily installed in any vehicle such as cars, boats and motorbikes. **EFY**



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